

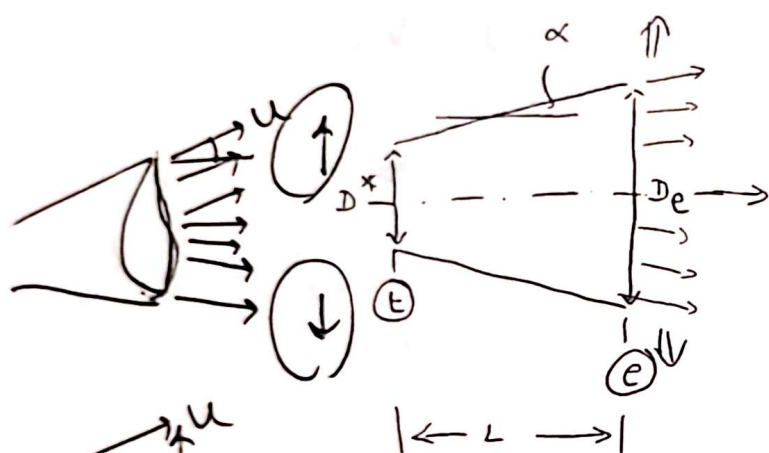
Nozzles, see 11.3, read pp 520-540.

Hill & Peterson
textbook

Diverging section nozzle shapes

Conical, perfect, maximum-thrust (aka minimum length), plug

Conical Nozzle, see fig 11.4



Exit flow is not axial
Some thrust is generated
in the transverse direction

not useful

$$\frac{D_e}{2} = \frac{D^*}{2} + L \tan \alpha$$

$$\Rightarrow D_e = D^* + 2L \tan \alpha$$

$$\frac{D_e}{D^*} = \left(1 + \frac{2L \tan \alpha}{D^*}\right) \quad \text{I gives L, obtain } D^* \text{ from I}$$

$$\frac{A_e}{A^*} = \left(\frac{D_e}{D^*}\right)^2 = \left(1 + \frac{2L \tan \alpha}{D^*}\right)^2 \Rightarrow L = \frac{D^*}{2 \tan \alpha} \left[\sqrt{\frac{A_e}{A^*}} - 1\right] \quad \text{II}$$

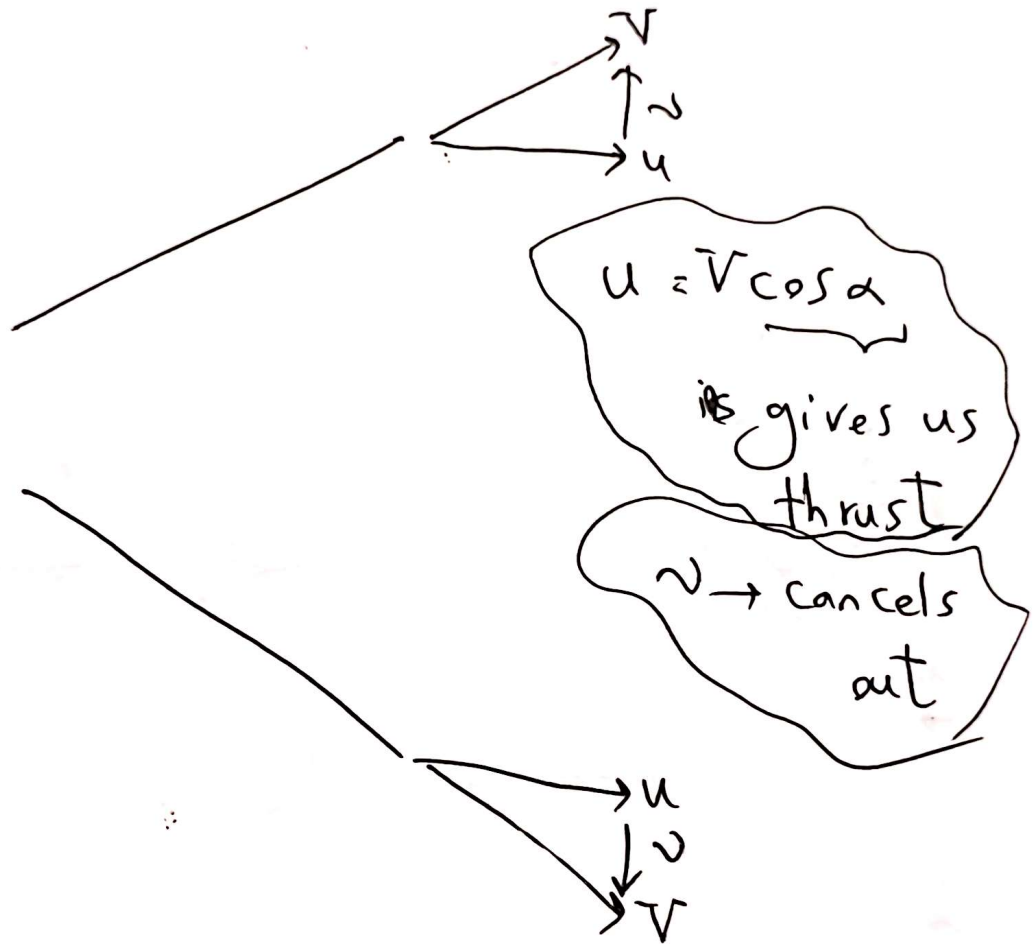
(see p 523)

gives L, obtain Ae from (2) Divergence loss for the conical nozzle (λ) give D^* & A_e , obtain L from III

$$\lambda = \frac{1 + \cos \alpha}{2} \quad (11.14)$$

Advantage: simple

Disadvantage: Exit flow is not completely axial. This introduces divergence losses (λ)



$$\alpha = 0 \quad \lambda = 1 \rightarrow \frac{1 + \cos \alpha}{2} = 1$$

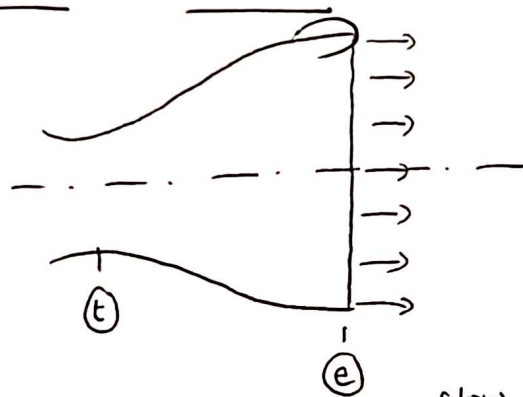
No loss



Comment: λ can be reduced by decreasing the nozzle half angle (α). This, however, results in a longer nozzle for a given area ratio (A_e/A^*).

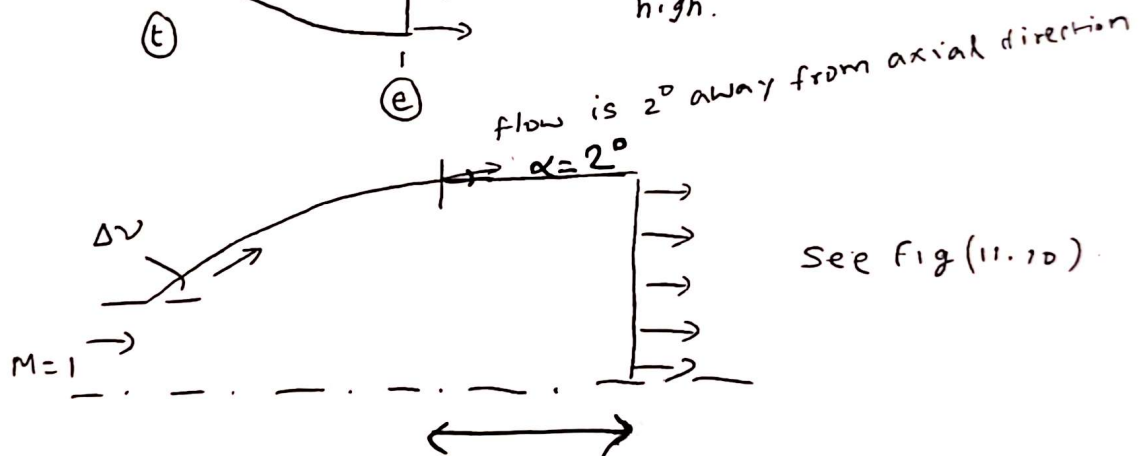
Perfect nozzle

Designed to achieve uniform, axial flow at exit.



Advantage: No divergence loss

Disadvantage: Very long.
The length needed to turn the flow the last few degrees is high.



See Fig (11.10)

Maximum thrust (or) minimum length nozzle

This nozzle is designed to generate the maximum Thrust for a specified length. See fig 11.13, and fig 11.14
(observe that the exit flow is not axial)

increasing $L \rightarrow$ reduces α

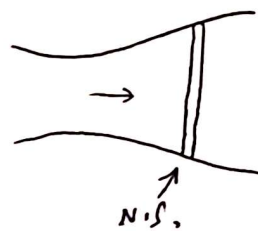
$$\frac{p_y}{p_x} = \frac{2\gamma}{\gamma+1} M_{xn}^2 - \frac{\gamma-1}{\gamma+1} \quad (0.5-)$$

$$\checkmark \quad \frac{p_a}{p_e} = \frac{2\gamma}{\gamma+1} M_{en}^2 - \frac{\gamma-1}{\gamma+1} \Rightarrow M_{en} = 2.135$$

$$M_{en} = M_e \sin \sigma \Rightarrow 52.6^\circ$$

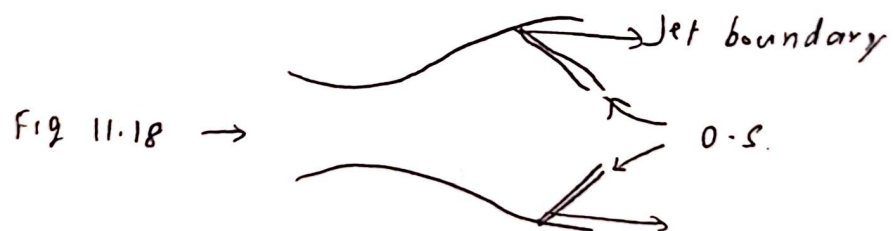
$\uparrow \qquad \qquad \uparrow$
 2.135 \qquad 4.25

Fig 11.18 shows a severely overexpanded case where the shock has moved into the nozzle.



$$P_0 A^* = \text{cte}$$

The actual scenario is the occurrence of 0.5.



Effect of back pressure

underexpanded jet: $p_e > p_a$

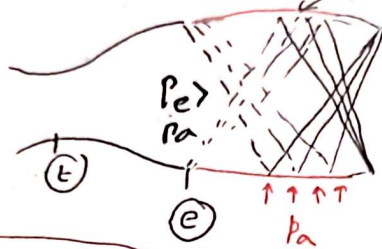
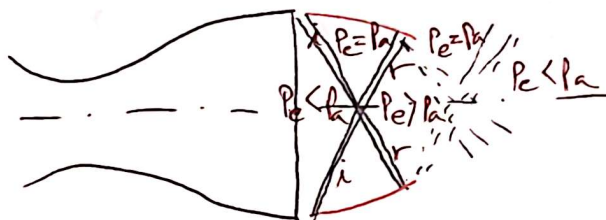


Fig 11.17 (a)

overexpanded jet: $p_e < p_a$



moderately overexpanded

Fig (11.17 b)

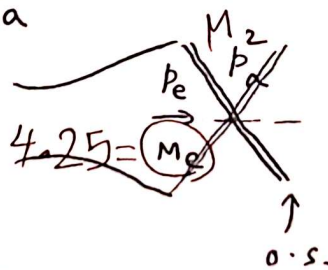
Severely overexpanded jet may result in shock moving into the nozzle.

[Ex: consider case 4 of Pbm 11.2

$$\frac{A_e}{A^*} \Rightarrow M_e = 4.25, p_e = 0.020476 \text{ MPa}, p_a = 0.1 \text{ MPa}$$

$p_e < p_a$

obtain oblique shock angle



$$\frac{p_a}{p_e} = 4.88$$

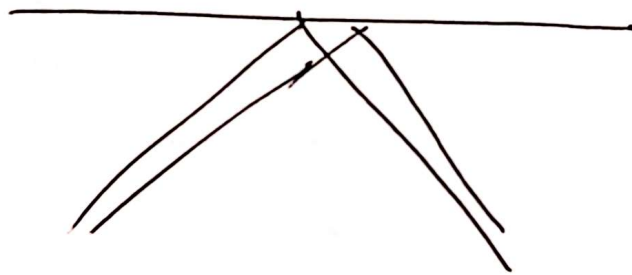


Solve this
Eq to get
 M_x

$$\frac{p_y}{p_x} = \frac{2\gamma}{\gamma+1} (M_x)^2 - \frac{\gamma-1}{\gamma+1} \quad (\text{N.S.})$$

$$2.135 = M_x^2 = M_e^2 \text{ in 6}$$

Can be obtained
4.25


$$f: \mathbb{R}^n \rightarrow \mathbb{R} \quad E_x \quad (5) \quad p = p_x$$

Conical Nozzles

Consider the flow out of the conical nozzle of Fig. 11.4, and assume that the streamlines in the expanding part of the nozzle are straight lines that all intersect at the point O . A control surface is indicated, passing through the spherical segment (of radius R) over which exhaust properties are constant. The reaction to the thrust is again shown as the vector $\mathcal{T}$. Applying the momentum equation (2.4), in the axial x -direction gives, for steady flow,

$$\sum F_x = \mathcal{T} + (p_a - p_e)A_e = \int_{cs} \rho(\mathbf{u} \cdot \mathbf{n}) u_x dA,$$

where A_e is the plane exhaust area πr^2 . No mass crosses the control surface except over the spherical exhaust segment, and there it is convenient to let

$$dA = 2\pi R \sin \phi R d\phi.$$

spherical cap

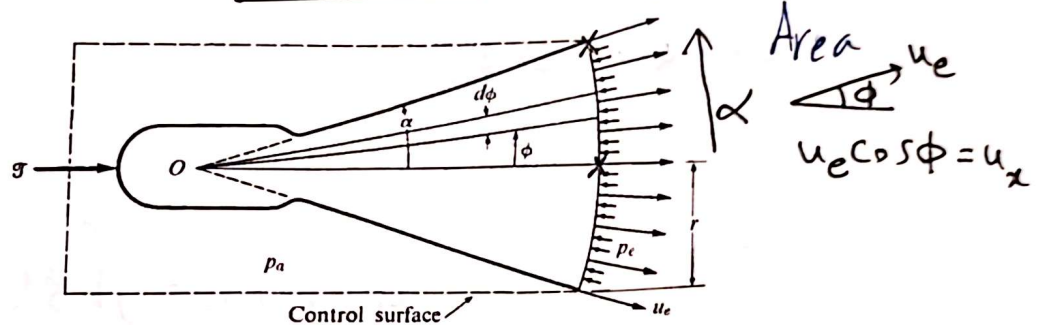
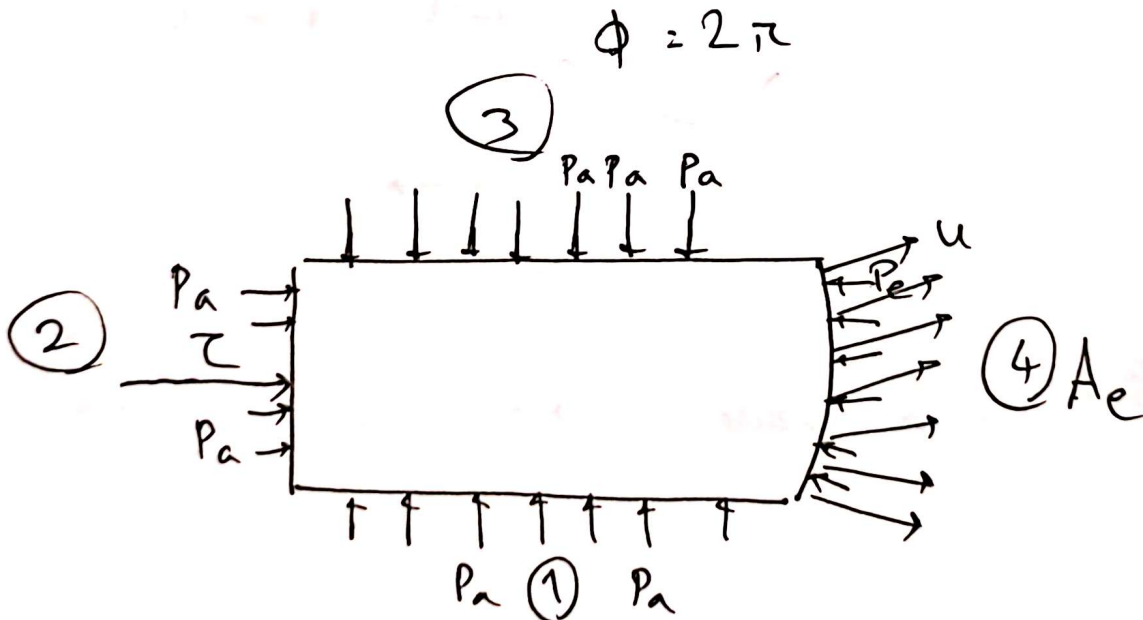
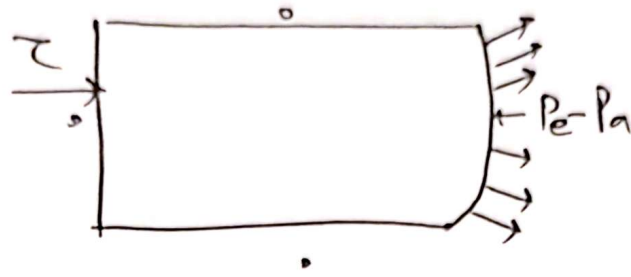


FIGURE 11.4 Spherically symmetric nozzle flow.



$$\tau + \int_{1-2-3} p_a dA + \int_4 p_e dA_e = \int \rho(\mathbf{u} \cdot \mathbf{n}) u_x dA$$

I want to get rid of this



I subtract p_a from all sides
because $\oint_S p_a dA = 0$.

$$\Rightarrow \tau + (p_e - p_a)A_e = \iint_S (\mathbf{u} \cdot \mathbf{n}) u_x dA$$

2-D Nozzle



for axial nozzle $\rightarrow$ R.H.S = $\underbrace{(\int u_x u_x dA)}_{\dot{m}} = \dot{m} u_x$

Further, in this case,

$$\underline{\mathbf{u} \cdot \mathbf{n} = u_e} \quad \text{and} \quad \underline{u_e = u_e \cos \phi}.$$

Thus

$$\mathcal{T} = \int_0^\alpha \rho(u_e) u_e \cos \phi 2\pi R^2 \sin \phi d\phi + (p_e - p_a) A_e.$$

Integrating, we obtain

$$(*) \quad \boxed{\mathcal{T} = 2\pi R^2 \rho u_e^2 \frac{1 - \cos^2 \alpha}{2} + (p_e - p_a) A_e.}$$

conical rocket thrust

Since the area of the spherical exhaust segment A'_e is $2\pi R^2(1 - \cos \alpha)$ the propellant flow rate is

$$\dot{m} = \rho u_e \times \frac{A_e}{2\pi R^2(1 - \cos \alpha)}, \quad \dot{m} = \int u_e A_e$$

and the thrust equation becomes

$$\boxed{\mathcal{T} = \dot{m} u_e \frac{1 + \cos \alpha}{2} + (p_e - p_a) A_e}$$

$$A_e = \iint_R dA$$

or, since the spherical area segment $A'_e = [2/(1 + \cos \alpha)] A_e$,

$$\boxed{\mathcal{T} = \frac{1 + \cos \alpha}{2} [\dot{m} u_e + (p_e - p_a) A'_e].} \quad (11.13)$$

Since, for reasonable nozzles, both the difference between A_e and A'_e and the contribution of the pressure term to the thrust are small, it is adequate to replace A'_e in Eq. (11.13) by the plane area A_e . Thus the thrust of an ideal conical rocket of half-angle α is reduced by a factor λ ,

$$\boxed{\lambda = \frac{1 + \cos \alpha}{2},} \quad \text{divergent loss of } (11.14)$$

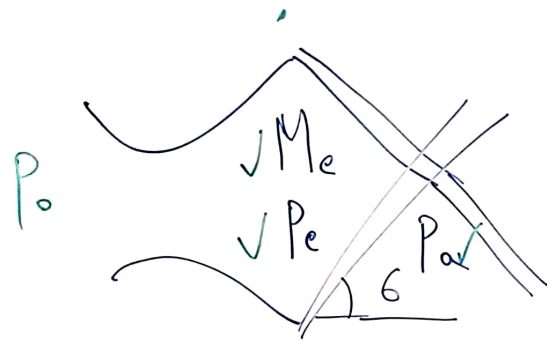
over that of an axial-outlet rocket with the same exhaust conditions.

conical nozzle

$$(**) \quad \tau = \dot{m} u_e + (p_e - p_a) A_e$$

Compare (*) with (**)

divide $\frac{1 + \cos \alpha}{2}$ to $\dot{m}$
you get



M_e is obtained from $\frac{A_e}{A^*} = f(M_e)$
 Sup.

$$\frac{P_e}{P_0} = \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{-\frac{\gamma}{\gamma-1}} \text{ gives } P_e$$

So you have $\frac{P_a}{P_0}$

$$\frac{P_a}{P_e} = \frac{P_2}{P_1} = \frac{2\gamma}{\gamma+1} \left(\overset{\text{known}}{M_e} \sin \delta \right)^2 - \frac{\gamma-1}{\gamma+1} \quad (\text{Shock-relation})$$

known Solve to get δ